Industrial Internship Report on

”Advanced Python Quiz Game”

Prepared by

Sandhya Omprakash Patel

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was (This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on developing a project titled "Advanced Python Quiz Game". The objective of this project was to design and develop an interactive quiz application using Python programming. The quiz game reads questions from a file and presents them to the user with different difficulty levels. It includes features such as timer-based answering, hint system, performance evaluation, and high score tracking. The project helped in understanding real-world problem solving and improved programming skills. It also provided hands-on experience in Python development and logical thinking. Overall, this internship was a great learning experience and helped in gaining practical knowledge of software development.) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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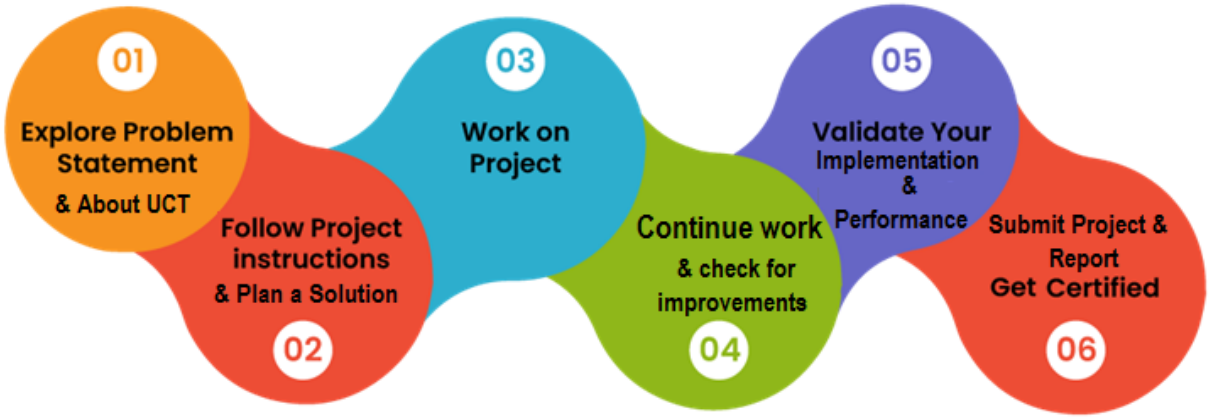
1 Preface

This report presents a summary of the work completed during the 6-week industrial internship conducted by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd.

The internship played an important role in understanding the need for practical exposure in career development. It provided an opportunity to apply theoretical knowledge to real-world problems.

During this internship, I worked on a project titled "Advanced Python Quiz Game", which involved developing an interactive quiz application using Python. The project focused on reading questions from a file, evaluating user responses, and displaying results.

The program was well-structured and included learning sessions, project development, and implementation phases. It helped me improve my programming skills, logical thinking, and problem-solving ability.



The above diagram represents the overall workflow of the system and shows how the quiz application processes user input and generates results.

I would like to express my sincere gratitude to my mentors, upskill Campus, and UniConverge Technologies for providing this valuable opportunity. Their guidance and support were very helpful throughout the internship.

This experience has motivated me to explore more advanced technologies in the future and continue improving my technical skills.

2 Introduction

This internship was conducted to provide practical exposure to real-world applications of Python programming.

During this internship, I worked on a project titled "Advanced Python Quiz Game". The objective of this project was to design and develop an interactive quiz application using Python.

The system reads questions from a file, presents them to the user, evaluates answers, and displays results. It includes features such as difficulty levels, timer-based answering, hint system, and performance analysis.

This project helped me understand the practical implementation of programming concepts and improved my problem-solving skills.

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

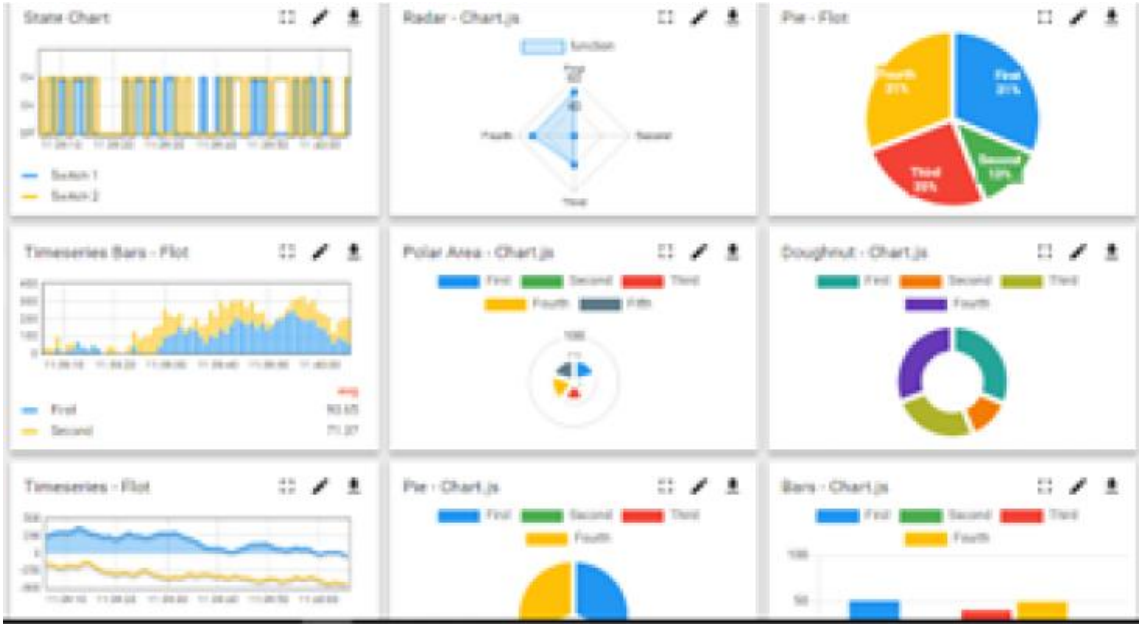
For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

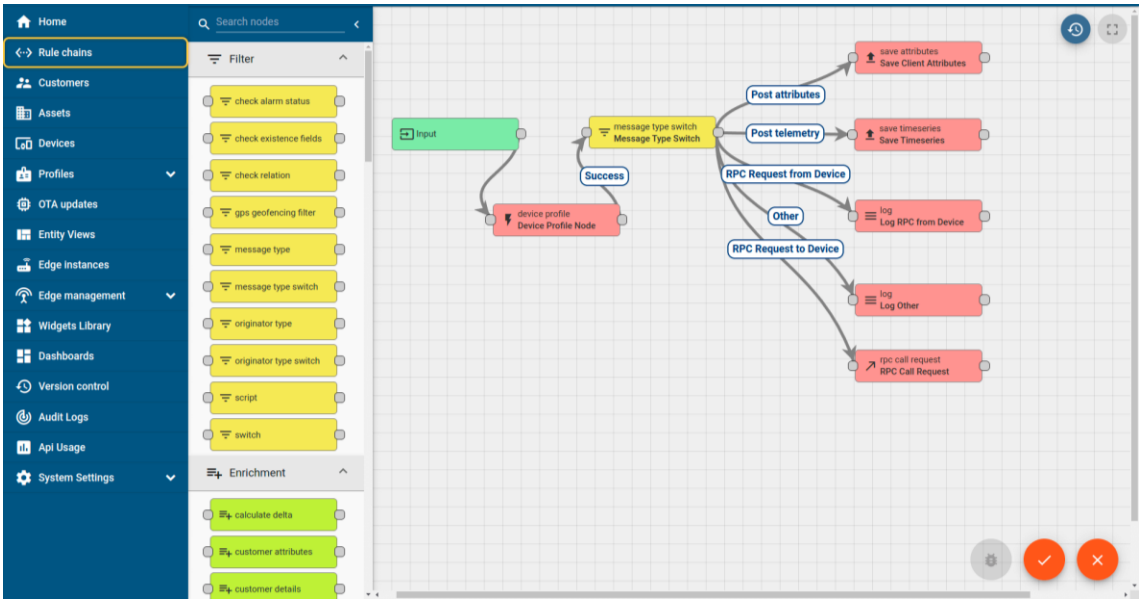
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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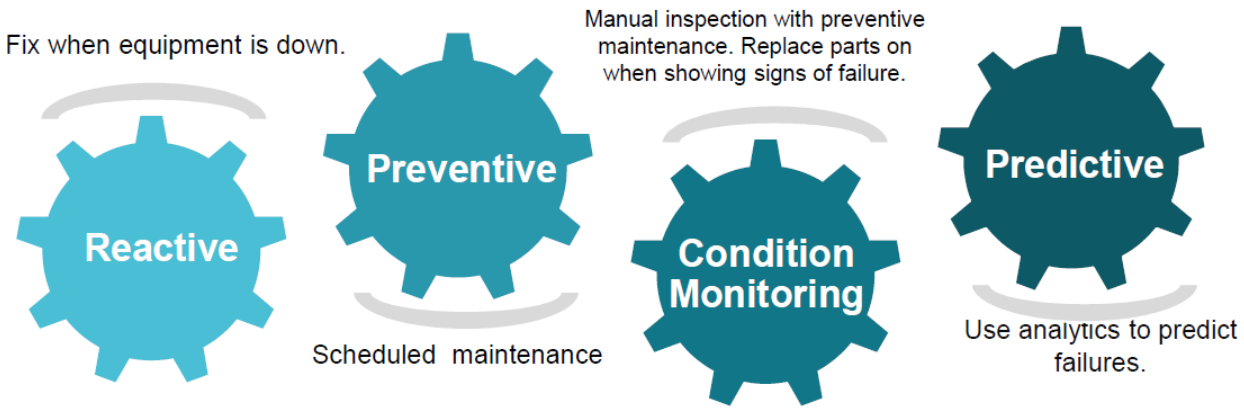
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

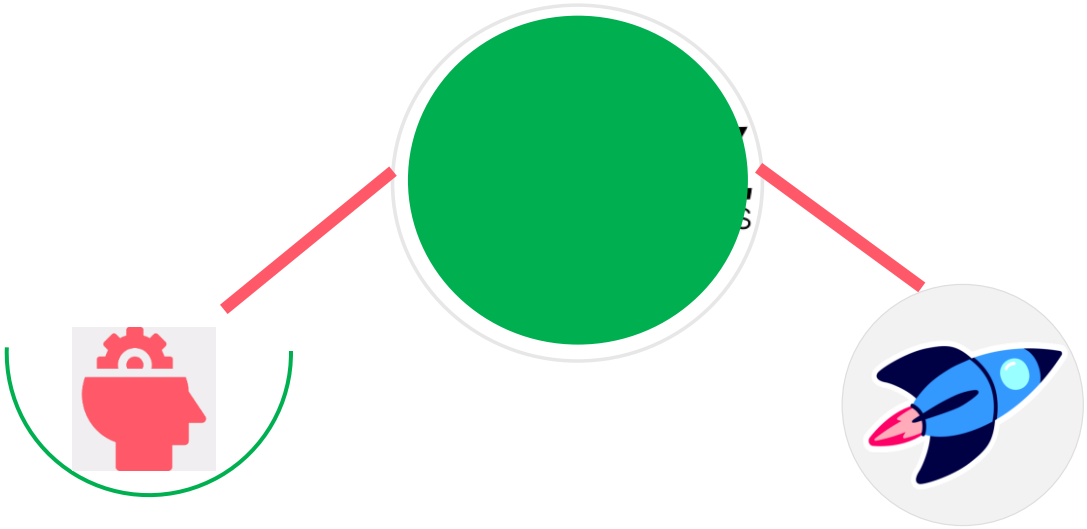
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

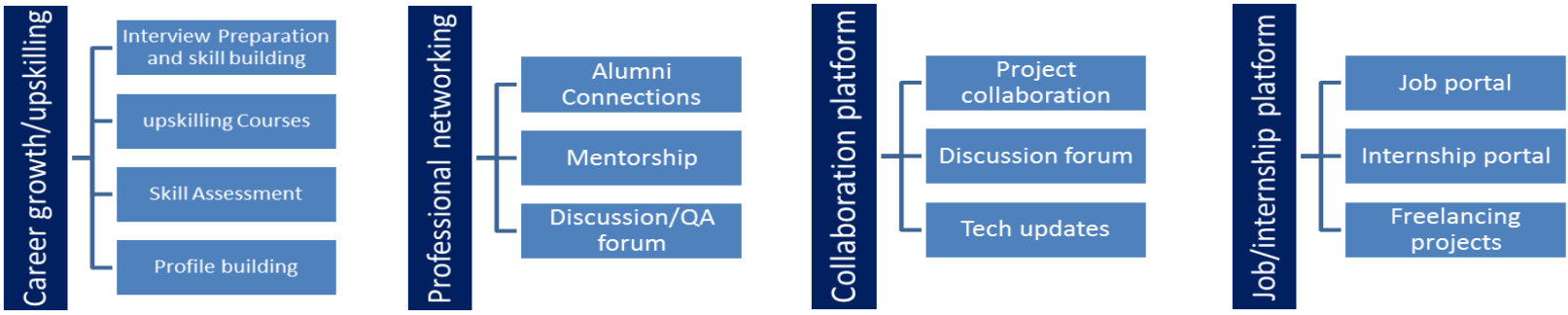
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

Additionally, this internship aimed to develop programming skills using Python and to gain hands-on experience in building real-world applications such as a quiz game system.

2.5 Reference

[1] Python Official Documentation – https://docs.python.org

[2] Online Python Tutorials – https://www.w3schools.com/python/

[3] Learning Resources from upskill Campus

2.6 Glossary

Terms	Acronym
1	Python – Programming Language
2	Quiz – Question-based evaluation system
3	File Handling – Reading/writing data from file

3 Problem Statement

The objective of this project is to design and develop an interactive quiz application using Python programming that can effectively test the knowledge of users.

Traditional quiz systems are often limited in functionality and use a fixed set of questions, which leads to repetition and reduced user engagement. There is a need for a system that can manage questions efficiently, provide a structured format, and offer a better user experience.

The proposed system reads questions from a file, presents them to the user based on selected difficulty levels, and evaluates the answers provided. It also includes features such as timer-based answering, hint system, score calculation, and performance analysis.

The main challenge was to design a user-friendly system that can handle multiple questions, process user input efficiently, and provide accurate results without errors.

This project aims to create an efficient and interactive quiz system that enhances learning and provides a better assessment platform for users.

4 Existing and Proposed solution

Existing Solution:

Traditional quiz systems generally use a fixed set of questions stored in a file or database. These systems are basic in nature and often lack advanced features such as difficulty levels, performance analysis, and user interaction. Due to the use of fixed questions, users may experience repetition, which reduces engagement and learning effectiveness.

Proposed Solution:

The proposed system is an Advanced Python Quiz Game that reads questions from a file and presents them to the user in a structured and interactive manner. The system allows the user to select difficulty levels such as Easy, Medium, and Hard.

The application displays questions one by one, accepts user input, evaluates the answers, and calculates the score. It also includes timer-based answering to increase challenge and improve user focus.

Value Addition:

- Implementation of difficulty levels (Easy, Medium, Hard)
- Timer-based answering system
- Hint system to assist users
- Score calculation and percentage display
- Rank-based performance evaluation
- High score tracking using file handling
- User-friendly interface and interaction

This proposed system provides a more efficient, interactive, and user-friendly quiz experience compared to traditional systems.

4.1 Code submission : https://github.com/Sandhyapatel-24/QuizGame.git

4.2 Report submission : https://github.com/Sandhyapatel-24/QuizGame/blob/main/QuizGame_Sandhya_USC_UCT.pdf)

5 Proposed Design/ Model

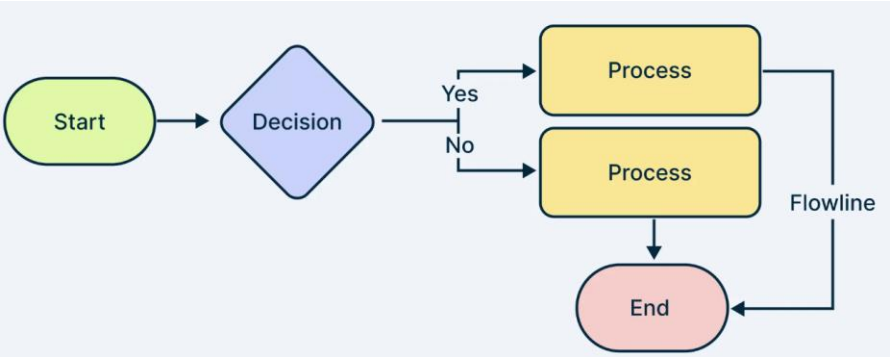
The proposed system is designed to provide an interactive quiz experience using Python programming.

The system follows a structured workflow:

1. Questions are stored in a text file.
2. The program reads the questions using file handling.
3. The user selects the difficulty level (Easy, Medium, Hard).
4. Questions are displayed one by one to the user.
5. The user enters answers within a given time limit.
6. The system evaluates the answers and calculates the score.
7. The final result and performance analysis are displayed.

This design ensures smooth interaction between the user and the system and provides an efficient quiz experience.

5.1 High Level Diagram :



User → Quiz Program → Read Questions File → Display Questions → User Input → Answer Evaluation → Result

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram

Load File → Store Questions → Loop → Take Input → Compare Answer → Update Score → Display Result

5.3 Interfaces

The system uses a command-line interface (CLI) where the user interacts with the program through input and output operations.

6 Performance Test

The performance of the quiz application was evaluated based on accuracy, speed, and user interaction.

The system was tested under different conditions to ensure smooth execution and correct functionality. Various test cases were used to verify the correctness of answers, timer functionality, and score calculation.

The application performed efficiently without any errors and provided accurate results. The response time was quick, and the system handled user input effectively.

6.1 Test Plan/ Test Cases

- Test Case 1: Valid input → Correct answer is accepted

- Test Case 2: Wrong input → Correct answer is displayed

- Test Case 3: Timeout → Question skipped after time limit

- Test Case 4: Level selection → Questions based on selected level

6.2 Test Procedure

- Run the program using Python

- Select difficulty level

- Answer questions

- Verify results and score calculation

6.3 Performance Outcome

- System works correctly

- Accurate score calculation

- Smooth execution

- User-friendly interaction

7 My learnings

During this internship, I gained valuable knowledge and practical experience in Python programming.

I learned important concepts such as file handling, loops, conditional statements, and user input handling. I also improved my logical thinking and problem-solving skills while developing the quiz application.

This project helped me understand how to design and implement real-world applications. It also improved my confidence in coding and debugging programs.

Overall, this internship was a great learning experience that will help me in my future career.

8 Future work scope

The project can be further improved by adding a graphical user interface (GUI) to make it more user-friendly.

Database integration can be implemented to store user data and scores permanently. Additional quiz categories can be added to expand the application.

The system can also be enhanced by implementing dynamic question generation and an online multiplayer quiz feature.

These improvements can make the application more advanced and scalable.